

Juan José Jáuregui Barba

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Education

Monterrey Institute of Technology and Higher Education (ITESM) Estimated graduation date: July 2026
Robotics and Digital Systems Engineering GPA: 3.5
Relevant Coursework: Intelligent Robotics, Control Systems, Embedded Systems, Robotics Foundation

Experience

Research Intern August 2025 – January 2026

National Institute of Advanced Industrial Science and Technology (AIST) Tokyo, Japan

- Developed robotic manipulation environments in Isaac Sim for reinforcement learning training and evaluation.
- Trained and evaluated reinforcement learning policies in Isaac Lab for Shadow Hand dexterous manipulation and UR5 grasping and reaching tasks across simulation and real-world scenarios.
- Deployed trained reinforcement learning policies from simulation to a real UR5 robotic arm using ROS 2 for real-world manipulation experiments.

Projects

Manchester Robotics – Warehouse AMR Autonomy Stack March 2026 – June 2026

Monterrey Institute of Technology and Higher Education (ITESM) Monterrey, Mexico

- Built a real-time C++ robot localization system with GPU-accelerated particle scoring, scan-to-map refinement, and loop closure running at 10 Hz.
- Implemented an ArUco localization layer using PnP pose estimation to rescue the MCL particle filter from drift in featureless corridors.
- Validated two complete autonomous pick-and-place missions on physical hardware, including real-time obstacle avoidance, pallet pick, and truck delivery end-to-end.

Manchester Robotics – Vision-Based Navigation March 2025 – June 2025

Monterrey Institute of Technology and Higher Education (ITESM) Monterrey, Mexico

- Developed ROS2 perception and control software for a Jetson-based autonomous robot.
- Implemented YOLOv8 sign detection, HSV segmentation, and multi-frame stabilization.
- Created a PID line-following controller and behavior-tree architecture for full mission coordination.

Robotic Arm Control and Motion Planning with ROS 2 January 2025 – February 2025

Monterrey Institute of Technology and Higher Education (ITESM) Monterrey, Mexico

- Developed a ROS 2 manipulation stack for a Moveo robotic arm using MoveIt 2 for motion planning and trajectory execution.
- Built and customized a parameterized URDF/Xacro robot model for simulation and visualization in RViz and Gazebo.
- Implemented motion planning workflows for robotic arm trajectory generation and manipulation tasks.

John Deere – Autonomous Waypoint Navigation August 2024 – November 2024

Monterrey Institute of Technology and Higher Education (ITESM) Monterrey, Mexico

- Developed a dual-core STM32 embedded system for autonomous waypoint navigation in agricultural vehicles.
- Implemented SPI, UART, I2C, CAN, and timer-based communication for real-time sensor and actuator integration.
- Integrated IMU and encoder data under an RTOS for state estimation and trajectory tracking control.

Skills

Programming: Python, C/C++, Rust, Bash, CUDA, Verilog

Technologies/Software: ROS 1/2, Isaac Sim, Isaac Lab, Gazebo, MoveIt, Docker, Linux, Git

Languages: Spanish (Native), English B2